

The University of Lahore, CS & IT Department

Object Oriented Programming

Assignment # 03

Start Date: 02/05/2025

Total Marks: 20

Due Date: 11/05/2025

Program: BSCS

Instructions

1. Understanding the problems is part of the assignment. So, no query, please.
 2. You will get zero marks if found any type of plagiarism or AI generated code.
 3. No submission after due date.
 4. Upload pdf file containing your code and screenshots of outputs.
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Q1. Mini-Facebook – Tree-Based Social Features.

You are tasked with building a simplified, console-based Mini-Facebook system using Binary Trees and Binary Search Trees. The system must manage a user's friend network and post feed using tree data structures, and provide basic social interactions like mutual friends and friend suggestions.

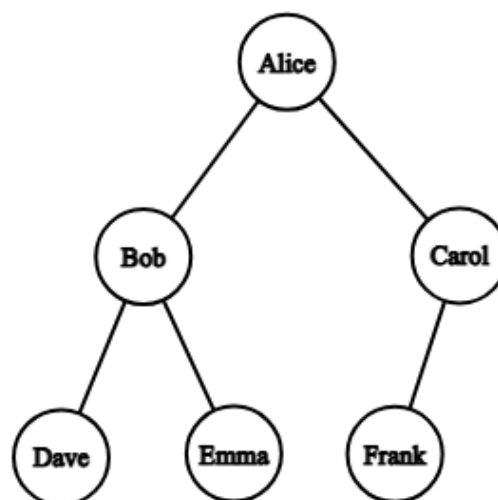
Feature 1: Friend Network (Binary Tree)

Each person can have up to two close friends: one on the left and one on the right (simplified assumption for binary tree structure). This forms a Binary Tree of friends.

Example Scenario:

Alice is the main user. She has two close friends: Bob (left child) and Carol (right child). Bob is also close to Dave and Emma, while Carol is friends with Frank.

This builds the following tree:



You have to implement the following functionality in C++:

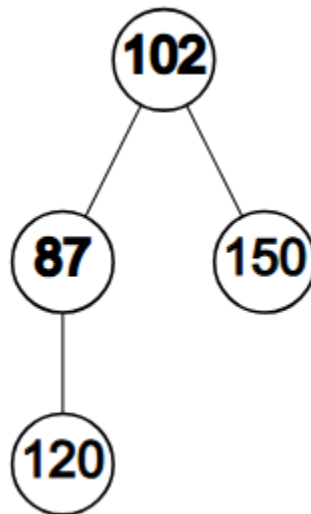
1. Insert a friend (as left or right child).
2. Delete a friend (simple or full deletion case).
3. Traverse the tree using:
 - Preorder – show how the network was built.
 - Inorder – list all friends alphabetically (if stored with names).
 - Postorder – for cleanup or deleting users.
4. Count total friends in the tree.

Feature 2: User Post Feed (Binary Search Tree)

Each user can post content with a unique Post ID (an integer). Posts are stored in a Binary Search Tree (BST) based on the Post ID to maintain sorted order.

Example Scenario:

Alice posts messages at different times having the IDs: 102, 87, 150, 120. The BST would arrange these to allow sorted access.



You have to implement the following functionality in C++:

1. Insert a new post by ID.
2. Delete a post by ID.
3. Display all posts using Inorder Traversal (to get them in order of creation).
4. Search for a specific post.

Feature 3: Mutual Friends (Binary Tree Traversal)

To find mutual friends between two users, traverse their respective friend trees and compare friend names.

Example Scenario:

Bob's friends: Dave, Emma, Grace

Carol's friends: Frank, Emma, Grace

Mutual friends = Emma, Grace

You have to implement the following functionality in C++:

1. Create two binary trees for two users.
2. Traverse each tree (e.g., Inorder) to extract friend lists.
3. Compare the lists and print mutual friends.

Feature 4: Friend Suggestions (Friends of Friends)

Suggest friends who are friends of your friends, but not directly connected to you.

Example Scenario:

Alice is friends with Bob and Carol.

Bob is friends with Dave and Emma.

Carol is friends with Frank.

Since Alice is not directly connected to Dave, Emma, or Frank, these names should be suggested.

You have to implement the following functionality in C++:

1. For a given user (e.g., Alice), get friends of their friends (one level deeper).
2. Exclude names already directly connected.
3. Print the list as friend suggestions.